

Komodo Dragon Element F

We were tasked with creating a box for Komodo dragons to be transported in and medically tested upon. Three groups in our class are working together to come up with one final product. Our group was tasked with creating the handles and determining the materials of the box. Our handles should be created in order to meet the following functional considerations: strength, comfort, adjustability, simplicity, cost, and repairability. We expect to have six handles on each box, for a total of eighteen handles, that satisfy our stakeholder's need for strength, comfort, foldability, and more. These features would make it easier to store in smaller spaces and better to carry around.

The design that we landed on for the handles was based on these functional evaluations and the weight we placed upon each function (the higher the weight, the more important the function). For example, criteria like strength, comfort, adjustability, and simplicity are weighted higher at 0.3, 0.2, 0.2, and 0.2 respectively. On the other hand, cost and repairability were weighted lower, both being 0.05. After all of our team members ranked each design - static handles, folding handles, static long bar, folding long bar, and divots - on a scale from one to ten, our design matrix computed final weighted scores for each design idea. The folding handles had the highest rating at 29.85, with the folding long bar at a close second with 29.7. This was how we eventually decided that folding handles was the best option.

For the box handles, adjustability/foldability were weighted as very important. On Thursday, January 30, 2025 we met with experts Nick Hanna, Steven Pfaller, James Flaherty, and Dr. Heather Schwartz at the Nashville Zoo. They explained that the handles folding down to fit right next to the box when not in use would be optimal. This advice and our prior knowledge from an online interview prompted us to weigh the foldability aspect higher than other criteria. Understanding their need for foldability for easy storage, we really only had two options for the main design of the handle: folding long bar and folding handles.



Figure 1F: Folding Long Bar



Figure 2F: Folding Handles

Another idea brought up by the experts at the Nashville Zoo was to put handles on both the sides of the boxes and at the ends. We hadn't originally planned to use handles on the ends, so this helped us narrow down the type of handle system to use. The ends of the boxes (20 inches wide) wouldn't be long enough to have a long bar on each end.

This, along with the fact that folding handles would take less materials to build than a long bar, basically gave us the one best solution for handles: folding handles.

However, as we progressed through designing the handle, a decrease in our original budget made it hard to afford everything that we had needed, so we got creative with the handles. We had to switch from 18 folding handles (6 on each box - two on each side and one on each end) to 6 folding long bars (two on each box). This was much more practical because it used less hinges and gave the used more space to grip the bar while carrying the Komodo Dragon. For the material of the long bars, we used spare PVC pipes that we found lying around and cut them down. We also used metal hinges lying around. We used twelve metal hinges and size long bars. These bars are all approximately 18 inches long and 1.25 inches wide. They will be placed on the sides (36 inches) of all three boxes.

Another task that our group was given was to determine the materials that would be used for the box. There are many design considerations that need to be taken into account. First of all, the materials need to be strong enough to hold up a Komodo dragon. The male Komodo dragon weighs approximately 330 pounds according to National Geographic. This means that each of the three boxes would need to be

able to support approximately 110 pounds if the weight of a Komodo dragon was distributed evenly. However, the Nashville Zoo experts told us that the male (largest) Komodo Dragon weighs approximately 200 pounds. This means that each box would need to carry about 67 pounds, which is a drastic drop from 110 pounds. We will still need to test the materials to make sure they meet the strength requirements. At first, we determined that we should make the box mainly out of wood. We were afraid that if we used polypropylene or polycarbonate that it wouldn't be strong enough to hold the Komodo dragon in if it got aggravated. At our meeting at the Nashville Zoo, they informed us that since the Komodo dragon would be confined in a small space (20 inches for both width and height), the dragons wouldn't have enough space to move to break through these types of materials as long as they were thick enough (1 inch). The zoo experts told us that this thickness of material would also be enough for the floors and ceilings of the box.

Other important design considerations are weather resistance, animal safety, weight, cost, and ease to work with. We created a design matrix to determine the materials needed. Using the design matrix, we determined the best materials to use. Our final design will incorporate aluminum, polycarbonate, and polypropylene, the three highest rated materials. In our time at the zoo, the experts explained that an aluminum frame would be optimal and polycarbonate could be used for the windows. This further solidified our design choice. The polypropylene could be used for the floors and walls that don't require windows.



Figure 3F - Aluminum

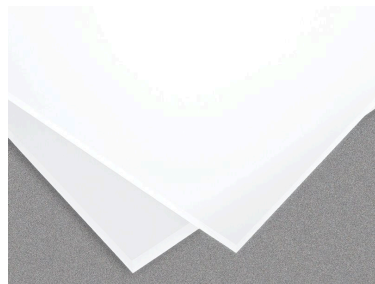


Figure 4F - Polypropylene



Figure 5F - Polycarbonate

Though we initially thought that wood would be the best material to use, we decided not to because of the expectations that it doesn't meet as well as aluminum does. For instance, this Komodo dragon cage will be outside for some time if the dragons are outside. This means the box needs to be weather resistant. Wood is probably the least weather resistant material that we considered. Strength and durability of material is also an important factor as expert Nick Hanna explained that they would want to use our final design forever. Wood would not last as long as aluminum in these situations. Aluminum is also generally stronger than wood also.

We were able to come in contact through another student in our engineering class with a company that might be able to provide the materials for the box creation for a low price. We asked them if they would be able to and it took them a week to decide on a price. About a week later, around mid March, they gave us their asking price: \$830. This was way over our budget of \$600. Everyone in our group would need to put in a bit of our own money if we wanted to make this work. We decided that we were going to use another plan: buy the parts ourselves and build the box. This would still be pricey though. We got together a cost breakdown of what we would need.

Aluminum Corners + Thicker Plastic

Aluminum angles - 12

- \$140 from home depot (w/ taxes, pick up is free)

<https://www.homedepot.com/p/Everbilt-3-4-in-x-3-ft-1-8-in-Thick-Aluminum-Angle-6405/332734017>

Polypropylene - 9 sheets

- \$18 per sheet (2 sheets per box)

<https://www.estreetplastics.com/1-8-x-24-x-36-Polypropylene-Sheet-Natural--p/9000122436.htm>

- \$32 per sheet (1 sheet per box)

<https://www.estreetplastics.com/Polypropylene-Sheet-Natural-1-4-Thick-p/x-900025.htm>

- Total: \$265 (w/ tax (exempt) + shipping)

Total Cost: \$405 materials cost

Need: 56 feet of metal

This was the plan we decided to go through with. We decided to use aluminum angles and polypropylene walls to build the frame of the box. This will cost us about \$400. This would give the other groups about \$200 to build a connection system and the medical insert areas.

Sadly, about midway through our project, we found out that our budget had almost been cut in half. Gratefully, we were able to get a friend in another engineering group, Corbin Wolfgang, to weld some steel together for us to make the frame. This cut the cost down so all we needed to order for the frame were the plastics: 1/4" thick plastics on the floor, 1/8" thick plastics for the walls, and clear plastics for the top. Once we drilled these into the final frame, our prototype of the frame of the box will be complete.

SOURCES:

<https://www.nationalgeographic.com/animals/reptiles/facts/komodo-dragon>